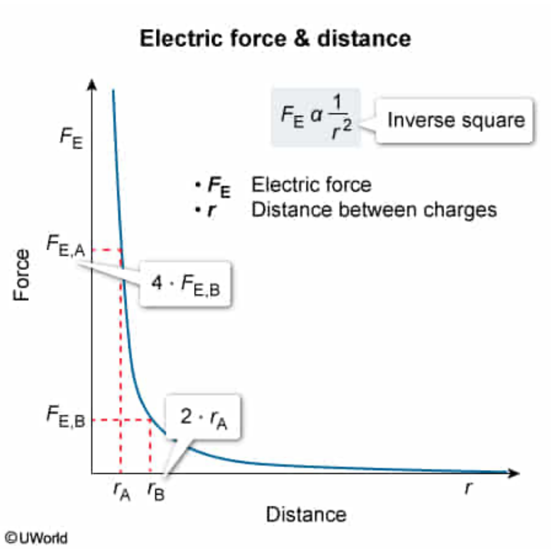
The electric force generated by two point charges separated by Distance A is 4 times greater than the electric force generated when separated by Distance B. How does Distance A compare to Distance B?

- A. Distance A is longer by a factor of 2. [9%]
- B. Distance A is longer by a factor of 4. [5%]
- ✔ C. Distance A is shorter by a factor of 2. [70%]
- D. Distance A is shorter by a factor of 4. [14%]

Correct
69% answered correctly

Explanation:



Just as every point charge generates an electric field, so any two point charges separated by some distance exert an **electric force (F_E)** on each other. Whether such forces are repellent or attractive depends on the sign of each charge, although all electric forces between charges are equal and opposite.

Coulomb's law relates the magnitude of the electric force generated by two point charges to the quantity of electric charge (q_1, q_2), the **distance (r)** separating the two charges, and the electric force constant (k = 8.99 × 10^9 N·m^2/C^2):

$$F_E = k \frac{q_1 q_2}{r^2}$$

Electric force always acts along an imaginary line connecting charges to one another in space. Accordingly, the general relationship between electric force magnitude and charge separation is contained within Coulomb's law:

$$F_E \propto \frac{1}{r^2}$$

Like the gravitational force between two objects, the relationship between the electric force and the distance separating each charge is an **inverse square function**. Consequently, tripling the distance between point charges decreases the electric force to 1/9th of the initial force.

In the question, the electric force generated by two charges separated by Distance A is 4 times greater than the electric force generated at Distance B. Stated differently, moving from Distance A to Distance B decreases the electric force by a factor of 4. According to Coulomb's law, this relationship would be observed when Distance A is shorter than Distance B by a factor of 2.

(Choices A and B) Coulomb's law dictates that Distance A must be *shorter* in length than Distance B if separating two charges by Distance A generates an electric force of greater relative magnitude.

(Choice D) Although the relationship between electric potential and distance is a first-power inverse relationship, Coulomb's law dictates that the relationship between *electric force* and distance is an *inverse square* relationship.

Educational objective:
Coulomb's law may be used to determine the electric force generated by two point charges separated by some distance. Electric force is inversely proportional to the square of the distance separating the two charges.

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